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A nonunit two-stage protection scheme for DC transmission lines in high-voltage DC grids

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ABSTRACT

Nonunit protection is suitable for the primary protection of DC lines in high-voltage DC (HVDC) grids because of its rapidity. However, most existing nonunit protections are not sensitive enough for remote-end high-resistance faults. To solve this problem, a novel nonunit two-stage protection capable of tolerating high-resistance faults is proposed in this paper. First, the propagation process of a traveling wave (TW) generated by the fault is presented, revealing the insufficiency of fault TW-based protections to tolerate remote-end high-resistance faults. Then, the TW generated by the DC circuit breaker (DCCB) located on the opposite end is analyzed, and its characteristic difference between the internal and external faults is discussed. Moreover, the nonunit two-stage protection scheme that contains stage-I and stage-II protections is proposed. Stage-I protection rapidly identifies the closed-end and remote-end low-resistance faults using the fault TW. Stage-II protection identifies the remote-end high-resistance fault using the TW generated by the DCCB, and the fault clearing time is reduced by combining the DCCB operation. The protection criteria, setting principle and operation process of the proposed protection scheme are presented in detail. Finally, a large number of PSCAD/EMTDC simulations validate the excellent performance of the proposed scheme, including a high fault resistance tolerance, quick fault clearing time, and no requirements of excessive sampling frequency and complex mathematical operations.

1. Introduction

The modular multilevel converter (MMC)-based high voltage DC (HVDC) grid has received widespread attention because it has technical advantages in terms of renewable energy integration, asynchronous grid interconnection, large capacity and long distance transmission [1–3]. However, once a DC fault occurs in HVDC grids, the system voltage drops significantly and the fault current rapidly rises [4], causing MMC blocking and equipment damage risks. To avoid the spread of the fault area and ensure equipment security, the DC fault zone should be identified and isolated within a few milliseconds, which imposes a high requirement for DC line protection [5]. Existing protection schemes for DC lines contain unit protection and nonunit protection [6]. Because two-end communication brings a significant delay, unit protection is usually used as backup protection for long transmission lines. Nonunit protection only uses the local electrical quantities to identify a fault and can be used as primary protection because the time delay of long-distance communication is avoided.

Conventional nonunit protections are mostly based on fault characteristics in the time-domain or frequency-domain. In [7], the rate of change of voltage (ROCOV) is used to identify the fault. Although this method is algorithmically simple, ROCOV is related to fault resistance and will lose selectivity in the case of high-resistance faults. In [8], the rate of change of current (ROCOC) is used to discriminate the fault type, but the identification of fault lines is not involved. Current limiting inductors are usually installed on both sides of lines to limit the fault current, and they can also provide a physical boundary for protection. In [9] and [10], the ratio of transient voltages at both sides of the inductor and the reactor power of the inductor are used to construct primary protections, respectively. However, these two primary protections cannot detect high-resistance faults. In [11], the change rate of the reactor voltage is used to detect faults, which is highly sensitive to double-pole faults (DPFs). However, this method requires a high sampling frequency and is not validated under pole-to-ground faults (PGFs). In [12] and [13], the high-frequency energy of line voltage is used to construct protections, but the high-frequency component decreases with the fault resistance. The above mentioned protection strategies have

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Nomenclature

DCCB	DC circuit breaker
MMC	modular multilevel converter
HVDC	high voltage DC
ROCOV	rate of change of voltage
ROCOLV	rate of change of line-mode voltage
N-PGF	negative pole-to-ground
DPFs	double-pole faults
P-PGF	positive pole-to-ground
TW	traveling wave

greatly promoted DC line protection for HVDC grids. Nevertheless, detecting high-resistance faults remains to be further researched for nonunit protections.

At present, there are two main methods to improve the protection sensitivity under high-resistance faults. For the first method, the fault characteristics that are rarely affected by the fault resistance are extracted from the fault traveling wave (TW) [14–17]. In [14], the ratio of high-frequency energy to low-frequency energy of line voltage is used to detect faults and achieves a high fault resistance tolerance. However, the time window of this method is up to 3 ms, which is unacceptable for closed-end serious faults. In [15], the first peak time of the line-mode component of fault voltage is used to identify a fault. In [16], a Levenberg-Marquart fitting method is used to extract the index coefficients from the zero-mode component of a fault TW to reflect the fault location. The methods in [15–16] are not affected by fault resistance but require a high sampling frequency. In[17], the arrival time difference between the zero-mode and line-mode components of a fault TW is used to identify the fault location; however, this method requires two-end communication to eliminate the protection dead zone and does not apply to double-pole faults (DPFs). For the second method, the TW generated by the DC circuit breaker (DCCB) operation is used to improve the protection sensitivity. The method in [18] uses the arrival time difference between the voltage TWs generated by the fault and the DCCB operation to construct a protection criterion; however, this method still has a small protection dead zone and requires a high sampling frequency.

In this paper, the TWs generated by the fault and the DCCB operation are used to construct a two-stage protection scheme. The proposed scheme uses local measurements to protect the whole transmission line, has good anti-interference ability and does not require excessive sampling frequency. Furthermore, the proposed scheme has good tolerance to fault resistance and can quickly clear the fault. Each section of the article is organized as follows. In Section II, the TW generated by the fault is discussed. In Section III, the generation and propagation of secondary TW under internal and external faults are analyzed. Using the characteristic difference between internal faults and external faults, a two-stage protection scheme that can tolerate high-resistance faults is presented in Section IV. In Section V, the performance of the proposed scheme is verified in a four-terminal MMC-HVDC grid. Finally, Section VI presents the conclusion.

2. Analysis of voltage TWs generated by the fault

As shown in Fig. 1, a four-terminal bipolar HVDC grid, which contains five transmission lines $l_1 \sim l_5$, is taken as an example to illustrate the TW process. Each converter station contains two MMCs, and the neutral point of each converter station is directly earthed. The lines are equipped with a current limiting inductor L_{dc} at both ends; R1 ~ R6 are the installation locations of the protection and hybrid DCCBs. In Fig. 1, faults f_1, f_3 and f_5 occur at lines l_4, l_1 and l_3, respectively; faults f_2 and f_4 occur at Bus1 and Bus4, respectively. In this paper, relay R1 is taken as the reference relay.

2.1. Modeling of the converter and transmission line

After a fault occurs, the sub-module capacitors in MMC discharge through the arm inductors. Before blocking, the MMC is equivalent to a series branch consisting of resistance R_{eq}, inductance L_{eq} and capacitance C_{eq} [6], and the total impedance Z_n is.

$$Z_n(s) = sL_{eq} + R_{eq} + 1/sC_{eq} \tag{1}$$

where s is the Laplace operator.

The above equivalent parameters are calculated by.

$$R_{eq} = 2r_0/3,\ L_{eq} = 2L_{arm}/3,\ C_{eq} = 6C_0/n_0 \tag{2}$$

where r_0 is the conduction resistance of the MMC arm, L_{arm} is the arm inductor, C_0 is the sub-module capacitor, and n_0 is the number of sub-module of the MMC arm.

To describe the TW phenomenon, the transmission line can be represented by the propagation function H(s) and surge impedance Z_c. In existing models, the frequency-dependent model is the most accurate and will be used in the simulation section; nevertheless, the fault calculation is not the core of this paper, and the Bergeron model is sufficient to describe the TW phenomenon in this section. Then, Z_c can be equivalent to a constant, and H(s) can be approximated as [16].

$$H(s) = \frac{K}{1+sT}e^{-sx/v} \tag{3}$$

where x is the propagation distance between the measurement point and the fault point, v is the propagation velocity, T is the dispersion time constant, and K is the ratio of the TW amplitude at the measurement point to the initial TW amplitude at the fault point.

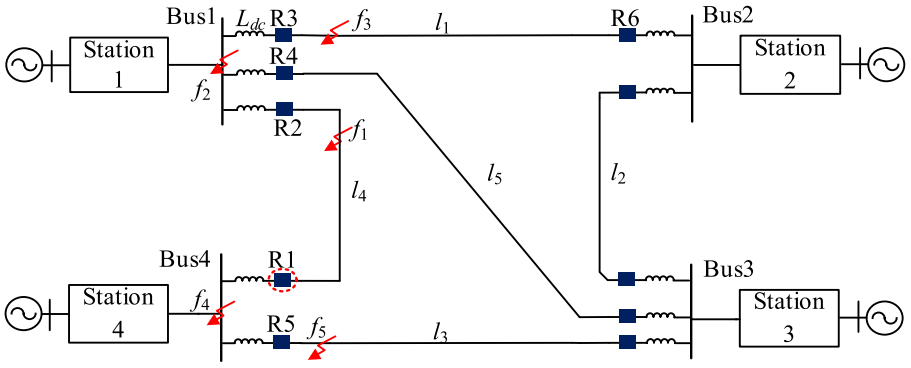


Fig. 1. Topology of a four-terminal MMC-HVDC grid.

2.2. Propagation process of TW generated by the fault

As shown in Fig. 2, after an internal PGF f_1 occurs, the initial TW at the fault location is marked as u_{in}. After an external PGF fault f_2 occurs, the TW propagates to line l_4 through the current limiting inductor, and the initial TW at the protection R2 is marked as u_{ex}. Considering that this section focuses on the characteristic difference of TWs generated by the internal and external faults, rather than the accurate fault calculation, the coupling of poles of transmission is ignored, and only the fault pole is considered in this section. During the internal and external faults, the initial TWs propagating to the protection R1, marked as $u_{R1,in}$ and $u_{R1,ex}$, respectively, are as follows:

$$\begin{cases} u_{R1,in}(s) = \alpha_0 H(s) u_{in}(s) \\ u_{R1,ex}(s) = \alpha_0 H(s) u_{ex}(s) \end{cases} \tag{4}$$

where α_0 is the refraction coefficient at location R1.

Before the refracted TW is reflected from Bus3 and the reflected TW is reflected from Bus1, both line l_4 and line l_3 can be equivalent to the wave impedance Z_c. Then, the equivalent circuit of location R1 is given in Fig. 3. The impedances on the right side and left side of R1 are $Z_R = Z_c$ and $Z_L(s) = sL_{dc} + Z_n(s)//(Z_c + sL_{dc})$, respectively. According to the refraction and reflection theorem, the expression of α_0 is.

$$\alpha_0 = \frac{Z_L(s)}{Z_R + Z_L(s)} \tag{5}$$

Similarly, equivalent circuits at internal and external fault locations can be established, which have been presented in [15] in detail and are not repeated here. Then, the expressions of $u_{in}(s)$ and $u_{ex}(s)$ are.

$$\begin{cases} u_{in}(s) = -\dfrac{Z_c}{2R_f + Z_c} \cdot \dfrac{U_{dc}}{s} \\ u_{ex}(s) = -\dfrac{2Z_c}{2Z_c + sL_{dc}} \cdot \dfrac{U_{dc}}{s} \end{cases} \tag{6}$$

where U_{dc} is the rated voltage of the HVDC grid.

As shown in Fig. 4, $u_{R1,in}$ and $u_{R1,ex}$ are calculated using (1)-(6). Referring to the simulation parameters in Section 5, the specific parameters are as follows: $K = 0.98$, $T = 1/40000$, $Z_c = 400\ \Omega$, $x = 500$ km, $U_{dc} = 400$ kV, $L_{dc} = 50$ mH, and $Z_n(s) = 0.12 + 0.013\,s + 1/(0.00045\,s)$. The TW generated by the external fault is significantly attenuated after passing through the inductor located at the remote-end. It has a significant smaller amplitude and decrease rate than the TW generated by the internal low-resistance fault. Thus, the first surge of fault TW can be used to distinguish the internal low-resistance fault and external fault. However, as given in (6), u_{in} weakens with increasing fault resistance. As shown in Fig. 4, when R_f is up to 300 Ω, the decrease rate of $u_{R1,in}$ in the time-domain and the amplitude of $u_{R1,in}$ in the frequency-domain are not larger than those of $u_{R1,ex}$. This fact means that the fault TW-based protection cannot be used to detect the internal high-resistance faults, and we use the TW generated by the DCCB operation to overcome this problem, which will be introduced in the next section.

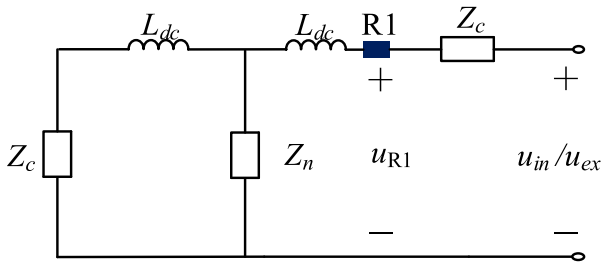


Fig. 3. Equivalent circuit of fault components under internal fault f_1 and external fault f_2.

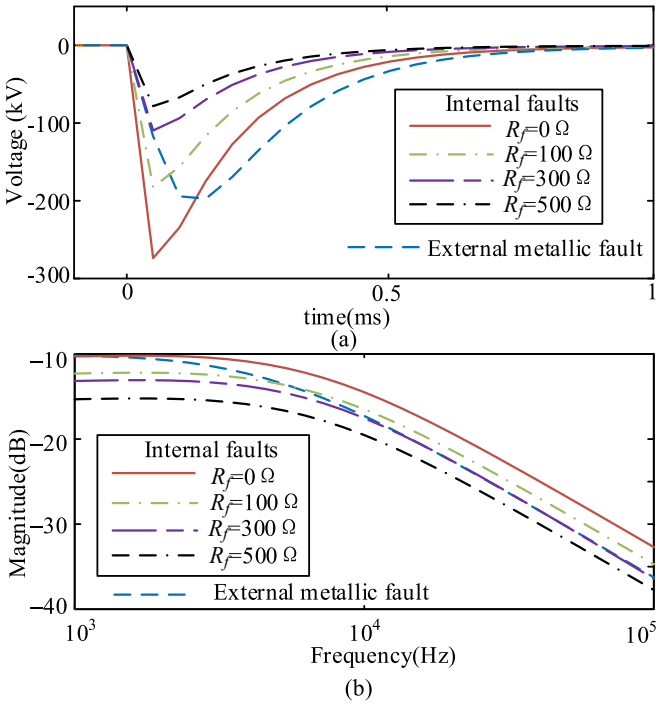


Fig. 4. First surge of the fault voltage TW: (a) waveforms in the time–domain; and (b) amplitude–frequency characteristics.

3. Analysis of voltage TWs generated by DCCB operations

3.1. Transient voltage generated by DCCB

Although DCCBs have different topologies, they all use arresters to absorb the residual energy of the transmission line, causing a large transient overvoltage on the arrester. The transient voltage propagates to other positions through the transmission line, resulting in a TW process. In this paper, the TW generated by the DCCB operation is called the secondary TW.

To illustrate the generation of secondary TW, it is necessary to

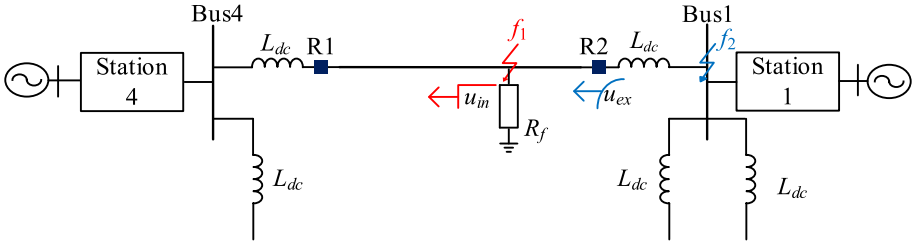


Fig. 2. Diagram of fault voltage TWs of an internal fault f_1 and external fault f_2.

introduce the DCCB operation process. As shown in Fig. 5, the typical DCCB presented in [19] is taken as an example, which is composed of the main conductor, main breaker and arrester, and the operation process is as follows.

(1) During the normal state, the system current flows through the main conductor with a rather small conduction loss.
(2) The first commutation: After receiving an interruption command, the IGBT modules in the main conductor are turned off, and the IGBT modules in the main breaker are turned on; thus, the fault current begins to be commutated from the main conductor to the main breaker. Once the fault current of the main conductor is 0 at $t = t_1$, the fast mechanical switch begins to open.
(3) The second commutation: Once the fast mechanical switch is totally opened at $t = t_2$, the IGBT modules in the main breaker are turned off, and the fault current is commutated from the main breaker to the arrester. Finally, the fault current gradually decreases to 0 at $t = t_3$. During t_2 and t_3, the secondary TW occurs and its maximum magnitude is limited to 1.5 times the rated voltage of the HVDC grid.

The polarity of a secondary TW depends on the fault current flowing through the DCCB. As shown in Fig. 6, for the reference protection R1, faults f_1, f_3 and f_5 are the internal fault, forward external fault and backward external fault, respectively. The DCCBs located at R2, R3 and R5 are used to interrupt faults f_1, f_3 and f_5, respectively. The transient overvoltages generated by the DCCB located at R2, R3 and R5 are denoted as U_{R2}, U_{R3} and U_{R5}, respectively. During these three interruptions, the fault currents of R2, R3 and R5 flow toward R1, backward R1 and backward R1, respectively. Therefore, the polarities of U_{R2}, U_{R3} and U_{R5} are positive, negative and negative, respectively, as shown in Table 1.

3.2. Refraction and reflection coefficient of line-mode component of TW

Considering the coupling between the positive and negative poles of a transmission line, the zero-mode and line-mode, which are decoupled from each other, are usually obtained as follows [20]:

$$\begin{cases} \begin{bmatrix} u_0 \\ u_1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} u_p \\ u_n \end{bmatrix} \\ \begin{bmatrix} i_0 \\ i_1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_p \\ i_n \end{bmatrix} \end{cases} \tag{7}$$

where u_p and u_n are the voltages of the positive and negative poles, respectively; i_p and i_n are the currents of the positive and negative poles, respectively; u_0 and u_1 are the voltages of the zero-mode and line-mode, respectively; and i_0 and i_1 are the currents of the zero-mode and line-mode, respectively.

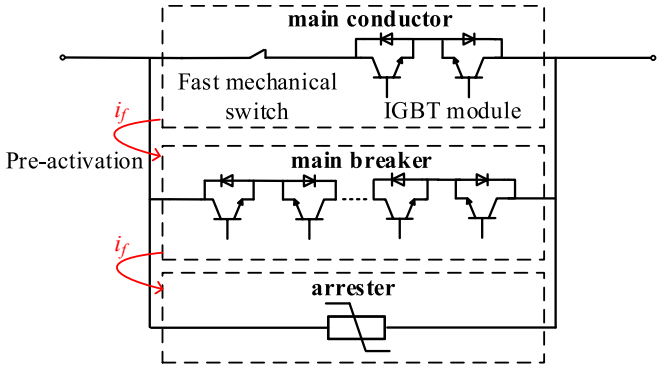


Fig. 5. Topology of the hybrid DCCB.

Considering the obvious dispersion of the zero-mode component, the line-mode component is used in the following analysis. When a line-mode TW u_x propagates from l_1 to l_4, the equivalent Peterson circuit is given in Fig. 7, where $2u_x$ is the Peterson voltage source, u_y is the line-mode TW at R2, Station 1 is represented by Z_n, and lines l_1, l_4 and l_5 are represented by the line-mode wave impedance Z_{c1}. When a line-mode TW u_x propagates from l_4 to Bus4, the equivalent Peterson circuit is the remaining circuit after removing the branch in the dotted box in Fig. 7, where u_z is the reflection line-mode wave at R1, Station 4 is represented by Z_n, and lines l_4 and l_3 are represented by Z_{c1}. Considering that the distance between the relay and nearby Bus is much smaller than the propagation distance of the TW during a sampling time interval, the multiple reflections between the relay and nearby Bus are neglected. By solving the equivalent Peterson circuits, the transfer coefficient α_1 when TW propagates from line l_1 to l_4 and the reflection coefficient β_4 at R1 are obtained as follows.

$$\begin{cases} \alpha_1 = \dfrac{u_y}{u_x} = \dfrac{2Z_nZ_{c1}}{Z_1(Z_1+3Z_n)} > 0 \\ \beta_4 = \dfrac{u_z}{u_x} = 1 - \dfrac{2Z_{c1}}{Z_1 + \dfrac{Z_nZ_1}{Z_n+Z_1}} > -1 \end{cases} \tag{8}$$

where $Z_1 = Z_{c1} + sL_{dc}$.

As shown in Fig. 8, a PGF fault occurs on the transmission line, and the fault resistance and the fault current are marked as R_f and i_{Fa}, respectively. When the line-mode TW u_{1f} propagates to the fault point, not only line-mode reflection wave u_{1b} and line-mode refraction wave u_{1r} but also zero-mode reflection wave u_{0b} and zero-mode refraction wave u_{or} occur at the fault point [18]. According to the boundary conditions and the converter between the phase-domain and mode-domain, we can obtain.

$$\begin{cases} u_{Fp} = i_{Fp}R_f \\ i_{Fn} = 0 \end{cases} \Leftrightarrow \begin{cases} u_{F1} + u_{F0} = (i_{F1} + i_{F0})R_f \\ i_{F1} = i_{F0} \end{cases} \tag{9}$$

where u_F and i_F are the voltage at the fault position and the current flowing through the fault resistance, respectively;

the subscripts 'p' and 'n' represent the variables of the positive pole and negative pole, respectively; the subscripts '1′ and '0′ represent the variables of the line-mode and zero-mode, respectively.

According to Fig. 8, we can obtain:

$$\begin{cases} u_{F1} = u_{1f} + u_{1b} = u_{1r} \\ u_{F0} = u_{0b} = u_{0r} \end{cases} \tag{10}$$

According to the Kirchhoff current theorem and Ohm's law in the mode-domain, we can obtain:

$$\begin{cases} i_{F1} = \dfrac{u_{1f}}{Z_{c1}} - \dfrac{u_{1r}}{Z_{c1}} - \dfrac{u_{1b}}{Z_{c1}} \\ i_{F0} = -\dfrac{u_{0b}}{Z_{c0}} - \dfrac{u_{0r}}{Z_{c0}} \end{cases} \tag{11}$$

where Z_{c0} is the zero-mode impedance of the transmission line.

Combining (9), (10) and (11), the refraction and reflection coefficients at the fault point, marked as α_f and β_f, respectively, can be obtained as follows.

$$\begin{cases} \alpha_f = \dfrac{u_{1r}}{u_{1f}} = \dfrac{Z_0 + 4R_f}{Z_1 + Z_0 + 4R_f} \\ \beta_f = \dfrac{u_{1b}}{u_{1f}} = \dfrac{-Z_1}{Z_1 + Z_0 + 4R_f} \end{cases} \tag{12}$$

As seen from (8) and (12), the values of α_f and α_1 are greater than 0, and the value of β_f is less than 0. Because $(k\text{-}1)Z_1$ is greater than Z_1, β_4 is greater than –1.

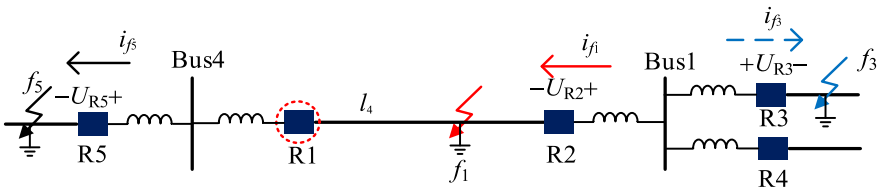


Fig. 6. Diagram of the protection location.

Table 1
Polarities of secondary TWs under different fault types.

Fault type	Polarity of the second TW	Current increment at R1
Internal fault f_1	Negative	Positive
External fault f_2	Positive	Positive
External fault f_5	Positive	Negative

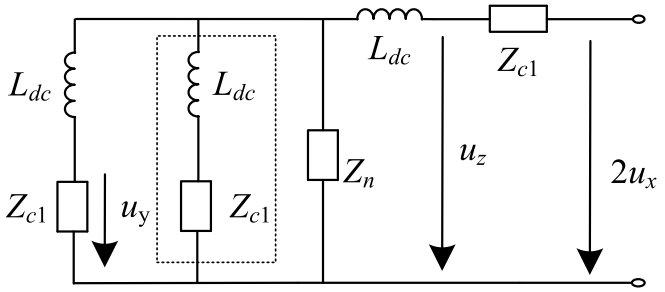


Fig. 7. Equivalent Peterson circuit at the end of the transmission line.

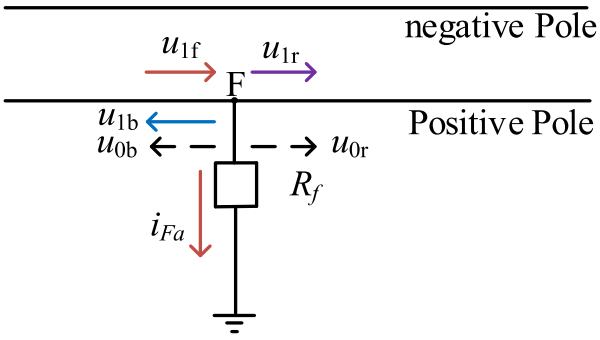


Fig. 8. Fault TW at the fault point.

3.3. Analysis of secondary TWs under Internal and external faults

(1) Analysis of the secondary TW under internal faults

The dead zone for conventional nonunit protection is the remote-end of the line. Nevertheless, a remote-end fault for local protection is close to the opposite protection and can be identified by the opposite protection. As shown in Fig. 6, although the local protection R1 may fail to identify the remote fault f_1, the opposite protection R2 can identify fault f_1. After a remote internal fault f_1 occurs, a secondary TW u_{in2} is generated by the DCCB at protection R2, and the TWs process between R1 and R2 is presented in Fig. 9.

As presented in the former section, u_{in2} has a negative polarity. The line-mode component of u_{in2} is denoted as $-U_0$. After receiving the first surge of the secondary TW, the change in line-mode voltage at protection R1 is denoted as ΔU_1. According to Fig. 9, ΔU_1 is calculated as follows:

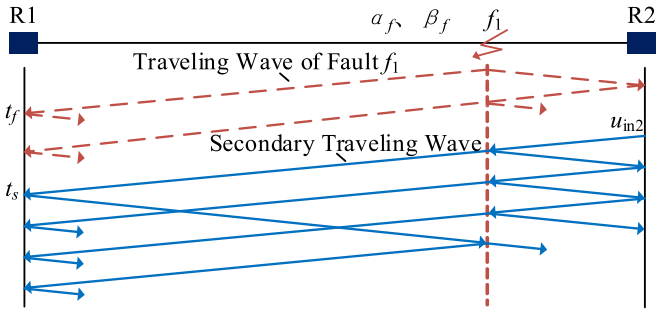


Fig. 9. TWs of a remote internal fault.

$$\Delta U_1 = -\alpha_f(1 + \beta_4)H(s)U_0 \tag{13}$$

The time for fault TW and secondary TW to reach R1, marked as t_f and t_s, respectively, is given in (14).

$$\begin{cases} t_f = \dfrac{x}{v_1} \\ t_s = \dfrac{l_{line} - x}{v_1} + t_I + t_{CB} + \dfrac{l_{line}}{v_1} \end{cases} \tag{14}$$

where v_1 is the propagation velocity of the line-mode TW, l_{line} is the transmission line length, t_{I} is the fault detection time of stage-I protection, and t_{CB} is the operation time of DCCB.

The time difference of t_f and t_s, marked as Δt, is.

$$\Delta t = t_s - t_f = \frac{2(l_{line} - x)}{v_1} + t_I + t_{CB} \tag{15}$$

(2) Analysis of the secondary TW under external faults

After an external fault f_3 occurs on the line adjacent to the local protection, the DCCB of the adjacent line operates rapidly, which generates a secondary TW u_{ex2}. The TWs after an external fault f_3 occurs are as follows:

As presented in the former section, u_{ex2} has a positive polarity. The line-mode component of u_{ex2} is denoted as U_0. After receiving the first surge of the secondary TW, the change in line-mode voltage at protection R1 is denoted as ΔU_2. According to Fig. 10, ΔU_2 is calculated as

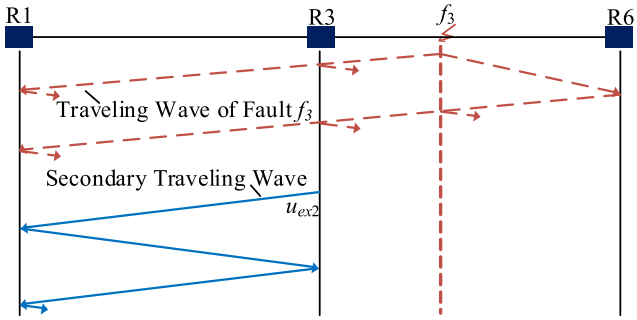


Fig. 10. TWs of an external fault.

follows:

$$\Delta U_2 = \alpha_1(1+\beta_4)H(s)U_0 \tag{16}$$

From (8) and (12), it can be seen that α_1, α_f, and $1+\beta_4$ are greater than 0. The propagation function H(s) only changes the amplitude of TW and does not change the polarity. Combining (13) and (16), ΔU_1 and ΔU_2 have opposite polarity. Using the simulation parameters in Section 5, the calculated results of ΔU_1 and ΔU_2 can be obtained. As shown in Fig. 11, the line-mode voltage at the local protection drops significantly and has a large decrease rate after receiving the first surge of the secondary TW of an internal fault, while it rises and does not have a large decrease rate after receiving the first surge of the secondary TW of an external fault. Therefore, local protection can identify internal remote-end high-resistance faults using the secondary TW.

4. Design of a nonunit two-stage protection scheme

4.1. Proposed protection scheme

(1) Protection triggering criterion

During the normal state, the local current i and voltage u are detected in real time. As given in (17), the triggering criterion is activated once ROCOC is larger than the threshold η.

$$\frac{di}{dt} > \eta \tag{17}$$

The selection of η should ensure that the triggering criterion is sensitive enough for internal high-resistance faults.

(2) Fault pole identification criterion

The magnitude of the zero-mode component of the fault voltage at the fault point is as follows [15]:

$$u_{f0} = \frac{-\sqrt{2}\mathrm{U}_{dc}Z_{c0}}{Z_{c0}+Z_{c1}+4R_f} < 0 \tag{18}$$

for a positive pole-to-ground fault (P-PGF), and.

$$u_{f0} = \frac{\sqrt{2}U_{dc}Z_{c0}}{Z_{c0}+Z_{c1}+4R_f} > 0 \tag{19}$$

for a negative pole-to-ground fault (N-PGF), and.

$$u_{f0} = 0 \tag{20}$$

for a DPF.

Therefore, the amplitude of the zero-mode component of the voltage at the local protection is used to identify the fault pole, as given in (21), and the operating time window of fault pole identification is denoted as T_3.

$$\begin{cases} \text{N - PGF} : u_0 > u_{set0} \\ \text{P - PGF} : u_0 < -u_{set0} \\ \text{DPF} : -u_{set0} < u_0 < u_{set0} \end{cases} \tag{21}$$

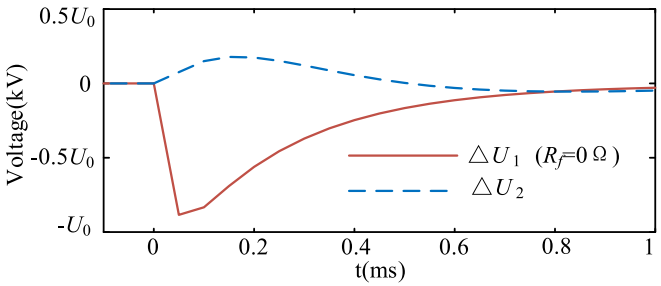


Fig. 11. Results of ΔU_1 and ΔU_2.

where u_0 is the amplitude of the zero-mode voltage during the time window T_3; u_{set0} is set to be greater than the maximum value of zero-mode voltage under the DPFs.

(3) Stage-I protection criterion

To identify metallic faults as soon as possible, the first surge of fault TW is used to construct stage-I protection. Because the inductor located at the line terminal weakens the TW generated by external faults, the values of ROCOV of internal metallic faults are greater than those of external faults [7]. The stage-I protection criterion is as follows:

$$\begin{cases} M = \left|\frac{du}{dt}\right| > u_{th1} = K_1 D \\ \Delta i = i - i_0 > I_{set} \end{cases} \tag{22}$$

where M is the absolute value of ROCOV during the time window T_1; u_{th1} is the threshold value; D is the maximum absolute value of the ROCOV during the time window T_1 for all possible external faults; K_1 is a reliability coefficient and can be set to 1.2; i_0 is the normal current value during the nonfault state; Δi is the current increment; and I_{set} is a current increment threshold.

For forward faults and backward faults, the values of Δi are positive and negative, respectively. Thus, I_{set} can be selected as a small positive value to avoid maloperation of protection in the case of backward faults.

(4) Stage-II protection criterion

As discussed in Section 3, the secondary TW of an internal fault causes the line-mode voltage to drop significantly. Thus, the rate of change of line-mode voltage (ROCOLV) is used to construct the stage II protection criterion, as given in (23), and the operating time window of stage-II protection is denoted as T_2.

$$\begin{aligned} &\textit{Under}\ \text{PGFs:} \begin{cases} P = \frac{du_1}{dt} < u_{th2} = K_2\sigma_1 \\ \Delta i = i - i_0 > I_{set} \end{cases} \\ &\textit{Under}\ \text{DPFs:} \begin{cases} P = \frac{du_1}{dt} < u_{th3} = K_2\sigma_2 \\ \Delta i = i - i_0 > I_{set} \end{cases} \end{aligned} \tag{23}$$

where P is the ROCOLV during the time window T_2; u_{th2} and u_{th3} are the threshold values designed for PGFs and DPFs, respectively; K_2 is a reliability coefficient and can be set to 1.5; and σ_1 and σ_2 are the minimum ROCOLVs during the time window T_2 for all possible external PGFs and DPFs, respectively.

The fault characteristics of DPFs are more obvious than those of PGFs, and stage-I protection is able to identify high-resistance DPFs. In this paper, stage-II protection is mainly used to improve the ability of tolerating high-resistance PGFs.

4.2. Protection operation process

In traditional protection logic, the DCCB interrupts the fault current after the protection has identified the fault, as shown in Fig. 12 (a). To reduce the fault clearing time, DCCB pre-activation is combined with the proposed protection scheme, as described below. As shown in Fig. 12 (b), once the protection triggering criterion is activated at $t = t_a$, indicating that the protection detects a potential internal fault, the DCCB begins to perform the first commutation. After completing the first commutation, the system current flows through the main breaker. The DCCB continues to perform the second commutation only after stage-I or stage-II protection identifies an internal fault at $t = t_b$. If stage-II protection identifies an external fault, the DCCB will return to its normal state by closing the fast mechanical switch and the IGBT modules in the main conductor.

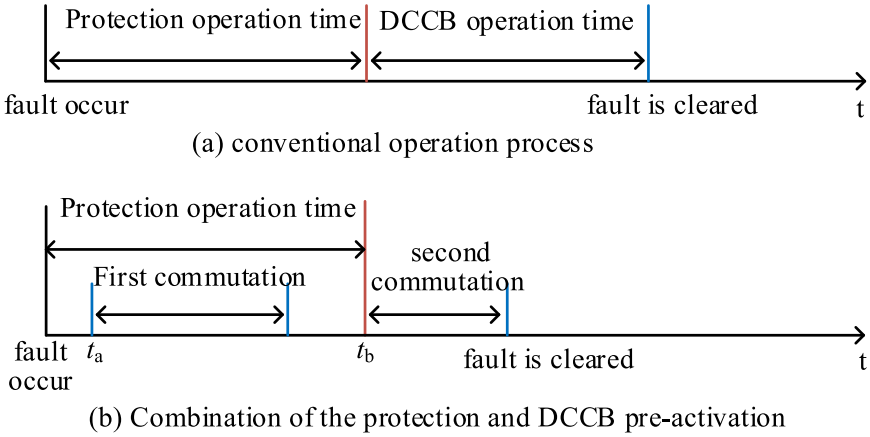


Fig. 12. Time sequence of the protection operation and DCCB operation.

As shown in Fig. 12, by combining the protection scheme and the DCCB pre-activation, the operation time of stage-II protection is overlapped with the DCCB operation time; thus, the fault clearing time is effectively reduced. Even if the DCCB does not have the pre-activation ability, the proposed protection scheme is still applicable; nevertheless, stage-II protection will require a longer fault clearing time.

The complete protection process is shown in Fig. 13. After an internal fault occurs, the triggering criterion in (17) is activated. The fault pole is identified using (21) during the time window T_3. Stage-I protection identifies the fault using (22) during the time window T_1. If the criterion in (22) is not satisfied, the time window of the stage-II protection is opened after a delay time of t_{delay}. Stage-II protection identifies the fault using (23) during the time window T_2. Once stage-I or stage-II protection identifies the fault, the DCCB on the fault pole performs the second commutation.

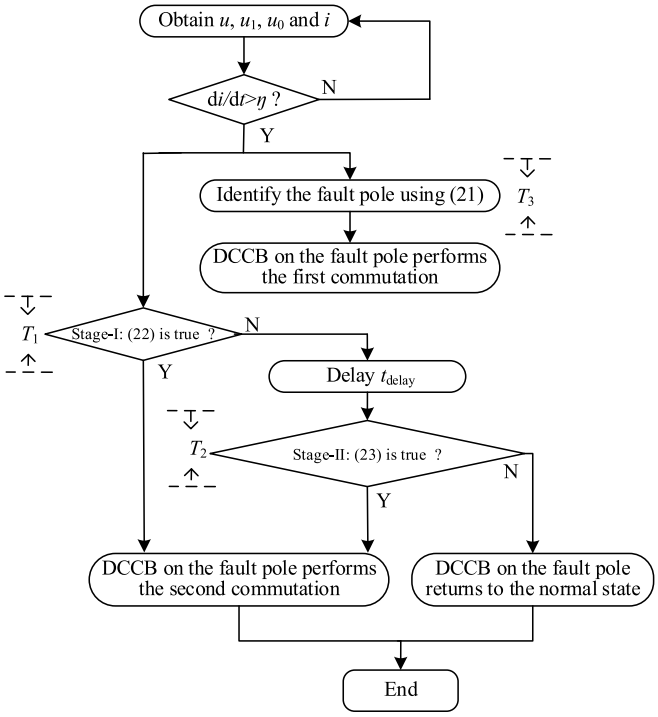


Fig. 13. Flowchart of the complete protection process.

4.3. Reliability of stage-II protection

After receiving the first surge of secondary TW of internal faults, the line-mode voltage significantly drops. During the time window T_2, the stage-II protection reliably detects internal faults using the ROCOLV, and it does not operate under other TWs, as explained below.

(1) For the TW generated by an external fault, the first surge is outside the time window T_2. Because of the attenuation effect of the transmission line, the subsequent refraction and reflection waves decay significantly, and the ROCOLV caused by these waves does not reach the threshold value of stage-II protection.
(2) For the secondary TW of a forward external fault, the arrival of the first surge causes a significant increase rate rather than a large decrease rate, and the subsequent refraction and reflection waves decay significantly. Thus, the ROCOLV does not reach the threshold value of stage-II protection.

5. Simulation and validations

A four-terminal bipolar ± 400 kV MMC-HVDC simulation model, as shown in Fig. 1, is built using the PSCAD/EMTDC platform, with the main parameters shown in Table 2. The overhead transmission line adopts a frequency-dependent model, and the detailed configuration of the transmission line and the tower structure is shown in Fig. 14. The half-bridge MMC is based on the equivalent model in [21]. For the DCCB, the time interval between the instant of receiving the interruption command and the instant of commutating the fault current into the

Table 2
Main parameters of the HVDC Grid.

Parameters	Values
Rated power of station (MW)	Station 1: −1000 Station 2: 1000 Station 3: 1000 Station 4: −1000
Length of the Overhead Lines (km)	line l_1: 200 line l_2: 500 line l_3: 500 line l_4: 500 line l_5: 500
Arm inductance L_{arm} (mH)	Station 1: 19 Station 2: 58 Station 3: 29 Station 4: 19
Capacitance of sub-module C_0 (μF)	Station 1: 450 Station 2: 150 Station 3: 300 Station 4: 450
Arm resistance r_0 (Ω)	Station 1: 0.18 Station 2: 0.55 Station 3: 0.27 Station 4: 0.18
L_{dc} (mH)	50
Number of arm sub-modules n_0	200
DCCB operation time (ms)	3
DCCB Interruption capacity (kA)	25
DCCB rated voltage (kV)	400 kV
DCCB peak interruption voltage (kV)	600 kV

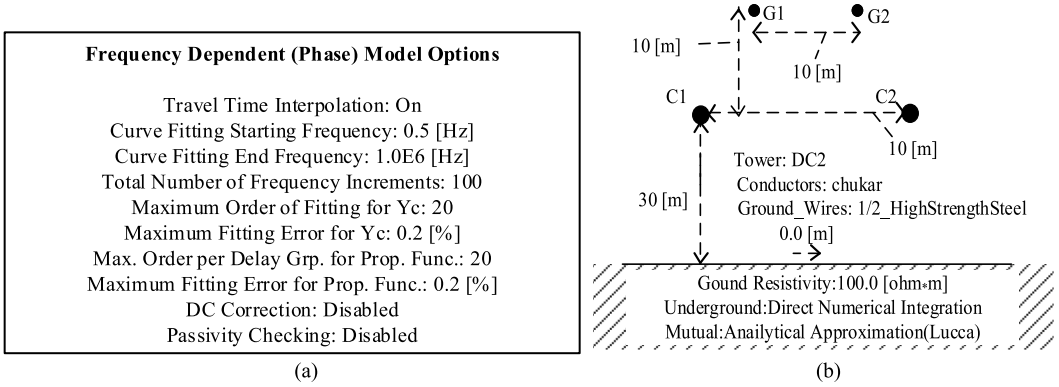


Fig. 14. (a) Detailed configuration of the transmission line; (b) tower structure.

arrester is called the operation time and taken as 3 ms. Referring to the DCCB in Zhangbei DC grid project [22], the main DCCB parameters used in this paper are given in Table 2, and the small stray inductor of the DCCB is ignored. The sampling frequency in this study is 50 kHz. Relay R1 is taken as the reference relay.

5.1. Setting of protection

(1) Setting of u_{th1}, u_{th2}, u_{th3}, u_{set0}, η and I_{set}: For protection R1, metallic fault f_2 on Bus1 is the most severe external fault. Under this fault condition, the D, σ_1, and σ_2 are 4556.28 kV/ms, −582.66 kV/ms and − 1399.16 kV/ms, respectively. Substituting K_1 of 1.2 and K_2 of 1.5 into (22) and (23), the threshold values u_{th1}, u_{th2} and u_{th3} are set to 5467.54 kV/ms, −873.99 kV/ms and − 2098.74 kV/ms, respectively. According to many simulations, the maximum value of the zero-mode voltage u_0 under DPFs is no more than 0.8 kV. The threshold value u_{set0} is taken as 5 kV to maintain a sufficient margin. In simulations, under a remote-end fault with 1 kΩ fault resistance, the value of di/dt is 1.12 kA/ms. To ensure that the triggering criterion can be reliably triggered when the fault resistance is up to 1 kΩ, the triggering threshold η is taken as 1.12 × 0.9 = 1.01 kA/ms. Considering that the current increment Δi is always positive under the forward fault and negative under the backward fault, we select I_{set} = 0.1 kA to avoid protection maloperation under backward external faults.

(2) Setting of T_1, T_3, T_2 and t_{delay}: To ensure that the arrival time of the first surge of fault TW is within the time window T_1, 100 μs before and after the triggering criterion activation is selected as the time window T_1, that is, the total length of T_1 is 200 μs. A large number of simulations have shown that equation (21) can select the fault pole within 50 μs after protection activation, and the time window T_3 is selected as 100 μs to maintain a sufficient margin.

According to (15), after the triggering criterion is satisfied at $t = t_f$, the secondary TW will reach the local protection R1 in the following time period (Δt_{min}, Δt_{max}), and t_f is the midpoint of time window T_1, as shown in Fig. 15. To reduce the risk of maloperation of stage-II protection, the stage-II protection time window T_2 should not be too long. Therefore, T_2 is not opened until a time delay of t_{delay}. Substituting simulation parameters into (15), when the fault distance x is equal to l_{line}, Δt has a minimum value of Δt_{min} = 3.1 ms. Since the stage-II protection is used to detect faults in the second half of the transmission line, the minimum value of x is $0.5l_{line}$ = 250 km, and the velocity of the TW is close to 300 km/ms; then, Δt has a maximum value of Δt_{max} = 4.77 ms. To ensure that the first surge of the secondary TW is within the time window T_2, ($T_1/2$ + t_{delay}) should be smaller than Δt_{min}, and ($T_1/2$ + t_{delay} + T_2) should be larger than Δt_{max}. In this paper, both t_{delay} and T_2 are set as 2.5 ms.

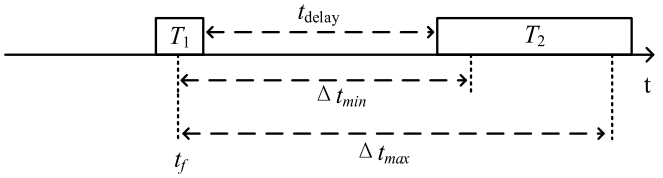


Fig. 15. Timing diagram of the protection operation.

5.2. The simulation results of internal faults

A metallic P-PGF f_{11} and high-resistance P-PGF f_{12} (R_f = 500 Ω) are set at the end of line l_4. The instant of fault occurrence is taken as 0 ms. The simulation results of protection R1 are shown in Fig. 16.

As shown in Fig. 16 (a), the voltage at protection R1 drops significantly after the arrival of fault voltage TW under fault f_{11}. As shown in Fig. 16 (b), M reaches the maximum value of 7693.95 kV/ms at 1.68 ms, which satisfies the stage-I criterion. As shown in Fig. 16 (c), the value of u_0 during T_3 is much smaller than $-u_{set0}$. Thus, the protection identifies

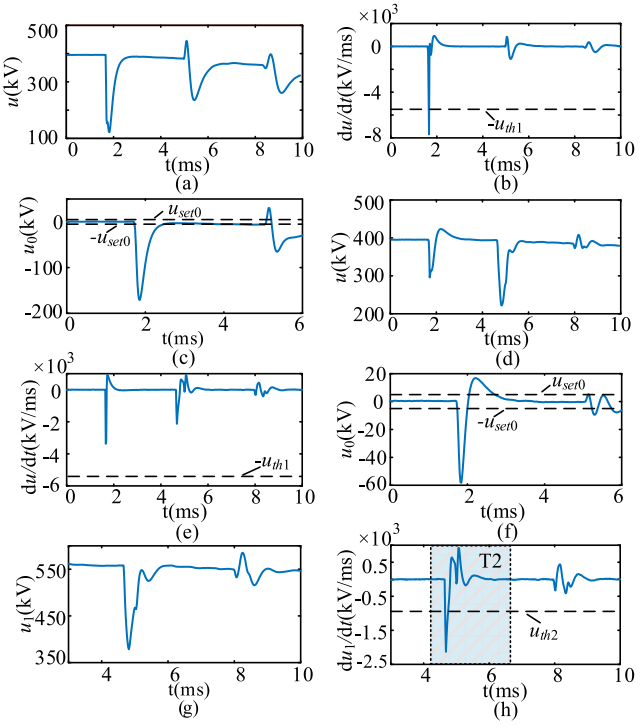


Fig. 16. Simulation results at protection R1:(a) voltage u under fault f_{11}, (b) du/dt under fault f_{11}, (c) zero-mode voltage u_0 under fault f_{11}, (d) voltage u under fault f_{12}, (e) du/dt under fault f_{12}, (f) u_0 under fault f_{12} (g) line-mode voltage u_1 under fault f_{12}, (h) du_1/dt under fault f_{12}.

fault f_{11} as an internal P-PGF.

As shown in Fig. 16 (d), the voltage at protection R1 drops after the arrival of the fault voltage TW under fault f_{12}. As shown in Fig. 16 (e), M reaches the maximum value of 3245.23 kV/ms at 1.68 ms, which does not satisfy the stage-I criterion. As shown in Fig. 16 (f), the value of u_0 during T_3 is much smaller than $-u_{set0}$. As shown in Fig. 16 (g)-(h), P reaches the minimum value of –2130.50 kV/ms at 4.7 ms, which satisfies the stage-II criterion. Thus, the protection determines that fault f_{12} is an internal P-PGF.

5.3. The simulation results of external faults

A metallic P-PGF f_2 and metallic P-PGF f_3 are set on Bus1 and the head of line l_1, respectively. The simulation results of protection R1 are shown in Fig. 17.

As shown in Fig. 17 (a)-(b), M reaches the maximum value of 2215.68 kV/ms at 1.68 ms, which does not satisfy the stage-I criterion. As shown in Fig. 17 (c)-(d), the time window T_2 is between 4.18 ms and 6.68 ms. P reaches the threshold value u_{th2} at 1.68 ms, out of T_2; thus, stage-II does not operate. During T_2, the minimum value of P is –582.66 kV/ms, which does not reach the threshold value u_{th2} of stage-II protection.

As shown in Fig. 17 (e)-(f), M reaches the maximum value of 426.71 kV/ms at 1.68 ms, which does not satisfy the stage-I criterion. As shown in Fig. 17 (g)-(h), T_2 is between 4.18 ms and 6.68 ms. During T_2, the minimum value of P is –193.34 kV/ms, which does not satisfy the stage-II criterion.

According to the simulation results, after the external fault, the fault TW and secondary TW will not cause maloperation of the stage-II protection because the time constraint and amplitude constraint in (23) are not satisfied simultaneously. This fact is consistent with the reliability analysis in Section 4.3.

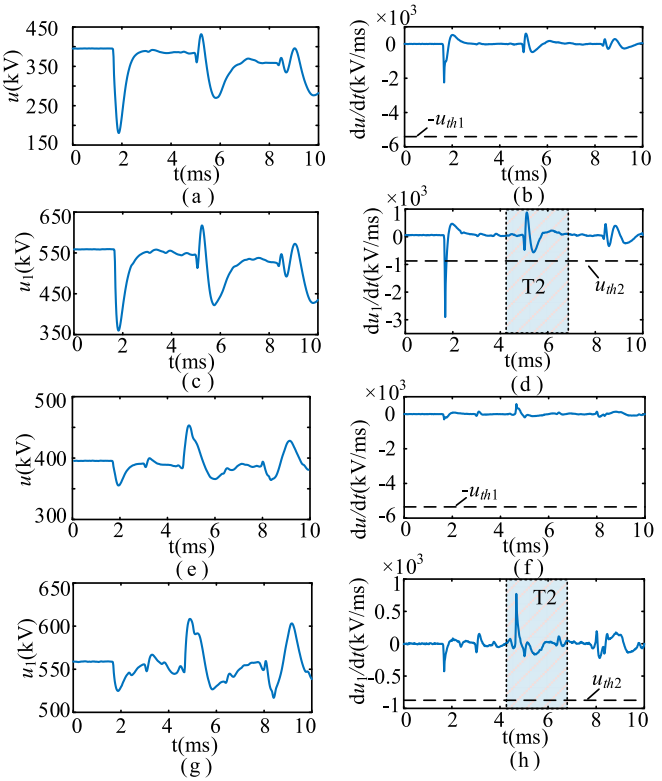


Fig. 17. Simulation results at protection R1:(a) voltage u under fault f_2, (b) du/dt under fault f_2, (c) line-mode voltage u_1 under fault f_2, (d) du_1/dt under fault f_2, (e) voltage u under fault f_3, (f) du/dt under fault f_3, (g) line-mode voltage u_1 under fault f_3, (h) du_1/dt under fault f_3.

5.4. Protection performance analysis

(1) Fault clearing time

The time from the fault occurrence to the beginning of the fault current decrease is defined as the fault clearing time. As shown in Table 3, faults with different locations and resistances are set in the HVDC grid. Stage-I protection can detect faults in the first half of the transmission line and low-resistance faults in the second half of the transmission line. Stage-I protection detects faults within 1.7 ms, and the fault clearing time is no more than 4.7 ms. Stage-II protection can detect high-resistance faults in the second half of the transmission line within 5.6 ms, and the DCCB then performs a secondary commutation to remove the fault current.

The fault clearing time of stage-II protection is only 0.9 ms later than that of stage-I protection because the operation time of DCCB overlaps the delay time window of stage-II protection. Therefore, DCCB pre-activation effectively improves the quickness of stage-II protection.

(2) The ability of tolerating fault resistance

The tolerable fault resistances of protection R1 are obtained using a large number of simulations, as shown in Fig. 18. The sensitivity of the stage-I protection decreases as the fault distance increases. Stage-II protection significantly improves the fault resistance tolerance capacity for faults in the second half of the transmission line, but it fails to improve the fault resistance tolerance capacity for faults in the first half of the transmission line. This fact is explained as follows. The local stage-II protection is based on the secondary TW generated by the DCCB located on the opposite end; thus, it actives only after the stage-I protection of opposite end actives. For faults in the second half of the transmission line, the stage-I protection of the opposite end tolerates a higher fault resistance than the local stage-I protection, indicating that the local stage-II protection can improve the fault resistance tolerance capacity. For faults in the first half of the transmission line, the stage-I protection of opposite ends tolerates a lower fault resistance than the local stage-I protection, indicating that the local stage-II protection cannot improve the fault resistance tolerance capacity.

As shown in Fig. 18, for a transmission line length of 500 km, the minimum value of tolerable fault resistance is 325 Ω, which occurs at the midpoint of the line. Stage-II protection can identify a remote end fault with a fault resistance of 800 Ω. Therefore, the proposed protection scheme has an excellent ability to tolerate fault resistance.

(3) Anti-noise interference ability

Using the simulation data in Sections 5.2 and 5.3, Gaussian white noise with a signal–noise ratio of 50 dB is added to the original measurements of internal high-resistance fault f_{12} and external metallic f_2, and the ROCOLV waveforms are shown in Fig. 19.

As shown in Fig. 19 (a), after the internal fault occurs, the minimum value of P during T2 is –2133.36 kV/ms, which satisfies the criterion of stage-II protection. As shown in Fig. 19 (b), after the external fault occurs, the minimum value of P during T2 is –663.55 kV/ms, which does not reach the threshold of stage-II protection. As shown in Fig. 19, although the noise deteriorates the waveforms of ROCOLV, the proposed protection can reliably distinguish internal faults from external faults.

When the noise is too great, the protection may fail to work, and noise reduction technologies, such as wavelet and mathematical morphology, are needed to assist the proposed protection in filtering out the noise. Nevertheless, this content is not the focus of this paper.

(4) Sampling frequency

According to the threshold value setting method in Section 4 and the system parameter in Section 5, the protection threshold values at

Table 3
Simulation results of protection R1.

Type	Fault Location	Distance (km)	Fault resistance (Ω)	Polarity of Δi	M (kV/ms)	P (kV/ms)	Identification result	Identification criteria	Fault clearing time (ms)
Pole-to-ground	f_4	0	0	Negative	/	/	External	/	/
			0	Positive	17371.51	/	Internal	Stage-I	3.34
		100	200	Positive	9230.18	/	Internal	Stage-I	3.34
			400	Positive	6284.47	/	Internal	Stage-I	3.34
	f_1		0	Positive	10508.36	/	Internal	Stage-I	3.84
		250	150	Positive	7295.26	/	Internal	Stage-I	3.84
			300	Positive	5566.51	/	Internal	Stage-I	3.84
			0	Positive	10778.155	/	Internal	Stage-I	4.00
		300	200	Positive	5717.55	/	Internal	Stage-I	4.00
			300	Positive	4630.31	−2912.92	Internal	Stage-II	5.34
			400	Positive	3891.68	−2987.72	Internal	Stage-II	5.34
			0	Positive	9277.24	/	Internal	Stage-I	4.34
			200	Positive	5257.03	−2859.40	Internal	Stage-II	5.02
		400	400	Positive	3711.51	−3260.68	Internal	Stage-II	5.02
			600	Positive	2813.78	−3479.85	Internal	Stage-II	5.02
			0	Positive	7693.95	/	Internal	Stage-I	4.68
			200	Positive	4725.35	−1884.26	Internal	Stage-II	4.70
		500	400	Positive	3440.68	−2157.48	Internal	Stage-II	4.70
			800	Positive	2285.50	−2822.89	Internal	Stage-II	4.70
	f_2	500	0	Positive	2215.68	−582.66	External	Stage-II	/
	f_3	500	0	Positive	426.71	−193.34	External	Stage-II	/
Double-pole		100	600	Positive	10471.05	/	Internal	Stage-I	3.34
	f_1	500	600	Positive	6729.59	/	Internal	Stage-I	4.66
		500	1000	Positive	4351.29	−5214.99	Internal	Stage-II	4.68

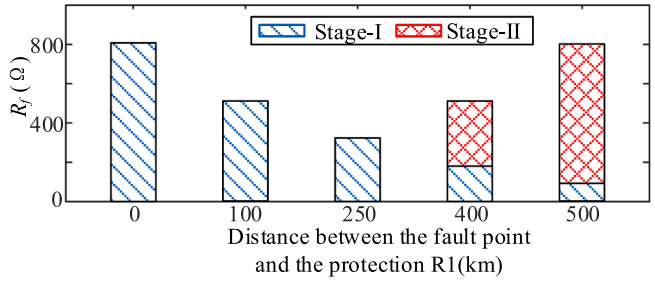


Fig. 18. The value of tolerable fault resistance of protection R1.

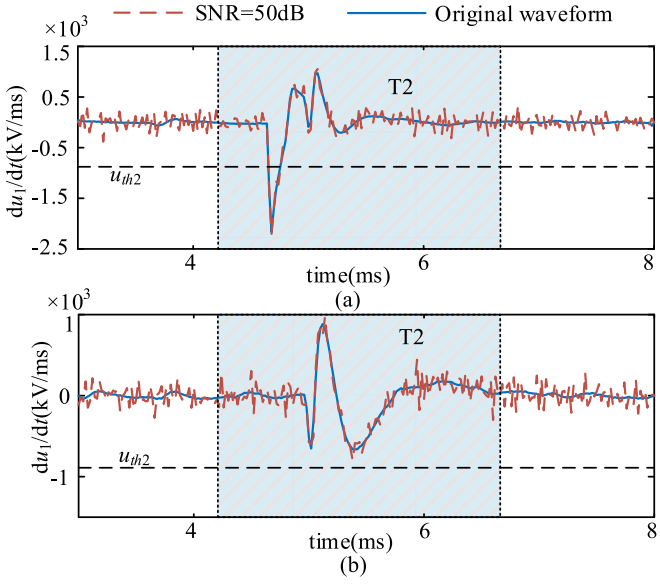


Fig. 19. Waveforms of ROCOLV under noise interference: (a) internal high-resistance fault f_{12}, (b) external metallic fault f_2.

Table 4
Protection threshold values at different sampling frequencies.

Sampling frequency	Threshold value of stage-I protection u_{th1}	Threshold value of stage-II protection u_{th2}
20 kHz	2433.02 (kV/ms)	−785.83 (kV/ms)
100 kHz	6226.95 (kV/ms)	−1015.79 (kV/ms)

different sampling frequencies are given in Table 4. We set a P-PGF fault at line l_4, and the fault distance is 500 km. The simulation results of R1 with different sampling frequencies are given in Table 5.

According to Table 2 and Table 5, when the sampling frequency is 20 kHz, 50 kHz and 100 kHz, the stage-II protection can detect fault resistances of 600 Ω, 800 Ω and 1000 Ω at a fault distance of 500 km, respectively. On the one hand, with the increase in sampling frequency, the sensitivity of protection increases because the protection can capture more TW information. On the other hand, the selection of sampling frequency is limited to the performance of available protection equipment. Nevertheless, compared with fault TW-based protection, stage-II protection can always improve the ability of tolerating high-resistance faults with different frequencies.

The simulation results are shown in Table 5.

(5) Current-limiting inductor

In this section, we take the current-limiting inductors as 30 mH and 100 mH, respectively. We set a P-PGF fault at line l_4, and the fault distance is 500 km. According to the threshold setting method in Sections 4, the protection threshold values are given in Table 6, and the simulation results of R1 are shown in Table 7.

According to Table 2 and Table 7, the fault resistance tolerance abilities of stage-I and stage-II protections increase with L_{dc}. This fact is explained as follows: 1) because L_{dc} suppresses the measured TW during the external fault, the fault resistance tolerance ability of stage-I protection increases with L_{dc}; 2) as discussed in Section 5.4.2, the fault resistance tolerance ability of local stage-II protection increases with that of opposite end stage-I protection. Compared with fault TW-based protection, stage-II protection can always improve the ability of tolerating high-resistance faults with different current-limiting inductors.

Table 5
Simulation results with different sampling frequencies.

Sampling frequency	Fault resistance(Ω)	M (kV/ms)	P (kV/ms)	Identification result	Identification criteria
20 kHz	0	2638.45	/	Internal	Stage-I
	100	1852.21	−856.29	Internal	Stage-II
	200	1418.39	−1170.01	Internal	Stage-II
	600	676.01	−1658.60	Internal	Stage-II
100 kHz	0	11509.35	/	Internal	Stage-I
	200	7627.41	/	Internal	Stage-I
	600	4551.69	−2059.14	Internal	Stage-II
	1000	3242.89	−2893.19	Internal	Stage-II

Table 6
Protection threshold values with different current-limiting inductors.

The value of L_{dc}	Threshold value of stage-I protection u_{th1}	Threshold value of stage-II protection u_{th2}
30 mH	7567.21 (kV/ms)	−1230.15 (kV/ms)
100 mH	3802.26 (kV/ms)	−717.15 (kV/ms)

Table 7
Simulation results with different current-limiting inductors.

L_{dc}(mH)	Fault resistance (Ω)	M (kV/ms)	P (kV/ms)	Identification result	Identification criteria
30	100	5467.91	−1354.47	Internal	Stage-II
	200	4363.34	−2149.50	Internal	Stage-II
	400	3098.43	−3051.34	Internal	Stage-II
100	200	5274.74	/	Internal	Stage-I
	400	3730.13	−1279.25	Internal	Stage-II
	800	2600.65	−1683.50	Internal	Stage-II
	1000	2223.84	−1790.48	Internal	Stage-II

5.5. Comparison with existing methods

In this section, relay R1 adopts the protections in [7] and [13], based on fault TW, to detect the remote-end fault on line l_4. The simulation results are given in Fig. 20, where u_{thr} and E_{thr} are the threshold values of protections in [7] and [13], respectively.

In [7], the internal fault is identified when the absolute value of ROCOV exceeds the threshold value, which is similar to stage-I protection. As shown in Fig. 20 (a), the absolute value of ROCOV decreases with the fault resistance, which is consistent with the analysis in Section 2. Once the fault resistance exceeds 100 Ω, the absolute value of ROCOV is smaller than u_{thr}, and the protection in [7] cannot detect the internal remote-end fault.

In [13], the energy of the high-frequency component of fault TW, marked as E_j, is used to detect the fault. As shown in Fig. 4, with the increase in fault resistance, the amplitude of the high-frequency component decreases; thus, the energy of the high-frequency component also decreases, which is validated by the simulation results in Fig. 20 (b). The protection in [13] improves the fault tolerance ability to 300 Ω. As shown in Table 3, the proposed scheme can identify the remote-end fault with a fault resistance of 800 Ω, indicating a better tolerance ability of fault resistance.

In addition, the proposed scheme only needs to calculate the change rate and increment of measurement values, which is mathematically simple; however, the wavelet transform and energy calculation are used in [13], which are relatively complex in mathematics. Thus, the proposed scheme has the advantage of the computational burden.

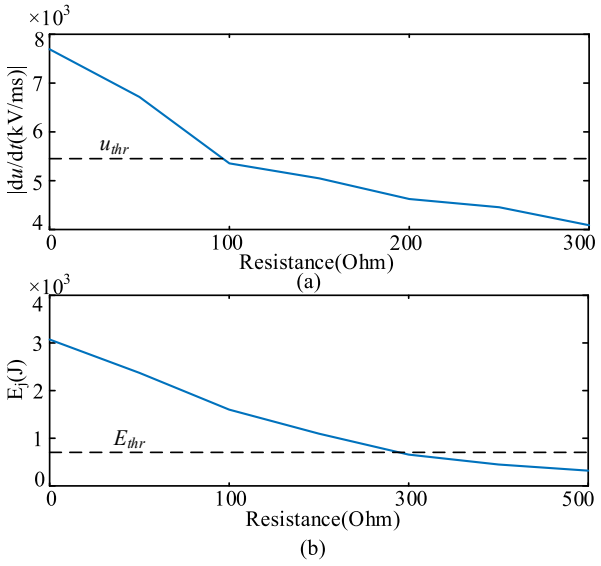


Fig. 20. (a) Relation of $|du/dt|$ and fault resistance; and (b) relation of E_j and fault resistance.

6. Conclusion

In this paper, a novel nonunit two-stage protection scheme capable of tolerating high-resistance faults is proposed for HVDC grids. The protection scheme distinguishes the fault and nonfault conditions using the rate of change of the current. Using the rate of change of voltage, stage-I protection detects the faults in the first half of the transmission line and the low-resistance faults in the second half of the transmission line. The characteristics of the secondary TW generated by the DCCB are analyzed. When the first surge of the secondary TW arrives, the line-mode voltage of local protection sharply decreases under an internal fault. According to this characteristic, stage-II protection detects the high-resistance faults in the second half of the transmission line using the rate of change of line-mode voltage. By combining DCCB pre-activation with the protection process, the quickness of stage-II protection is effectively improved.

The proposed protection scheme uses local measurements to detect transmission line faults, achieving a high fault resistance tolerance and not requiring long distance communication. In addition, the protection achieves quick fault clearing time and has no requirements of excessive sampling frequency and complex mathematical operations. These protection performances are conducive to the safety of HVDC grids.

CRediT authorship contribution statement

Yanxun Guo: Conceptualization, Methodology, Writing – original draft. **Zhuang Xu:** Writing – original draft, Visualization. **Yaoqiang Wang:** Writing – review & editing, Supervision. **Xiaomei Yao:** Writing – review & editing. **Jun Liang:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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